Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Fall, Year:2023), B.Sc. in CSE (Day)

Lab Report NO # 01
Course Title: Engineering Drawing
Course Code: CSE-208 Section: 221-D2

Lab Experiment Name: Implementation of 2D drawing.

Student Details

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Lab Date : 20-10-23
Submission Date : 24-10-23
Course Teacher's Name : Md Naimul Pathan

Lab Report Status

Marks: …… **Signature:.....**
Comments:..... **Date:.....**

1. TITLE OF THE LAB EXPERIMENT

Implementation of 2D drawing.

2. OBJECTIVES

i. It will help us to gain knowledge about 2D or 3D drawing platform Auto cad and how to use this software.

ii. We can learn about 2D drawing technique and uses of line, rectangle, circle, etc.

iii. In this experiment we will implement a provided figure using 2D drawing technique.

3. TOOLS AND TECHNOLOGIES

1. line:

Line command draws a straight line from one point to another.

Shortcut Key: L

2. Circle:

Circle helps us to draw a circle. We can draw circle in many ways like using radius, diameter, tan-tan radius, etc.

Shortcut Key: C

3. Trim:

It is used for cutting unwanted line and edges.

Shortcut Key: Tr

4. Ellipse:

Ellipse helps us to draw a Ellipse. We can draw Ellipse in many ways like using radius of both side, using diameter and half ellipse using different point.

5. Angle:

Angle helps us to draw line using different angle. To implement this first we have to select line then press 'Shift+@' and give line diameter then press '<' give angle (45) and then press 'Enter'.

6. Offset:

Offset helps us to create same line in a particular disance.

4. IMPLEMENTATION

1. Draw a line and make an offset line using distance between 95cm. Identify the distance in main line and mark it. Then draw a line and draw two ellipse using intersect point as center.

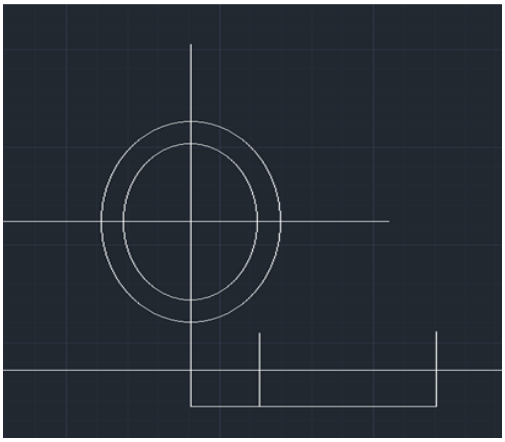


Figure : 01

2. Draw another Ellipse and a one at the bottom.

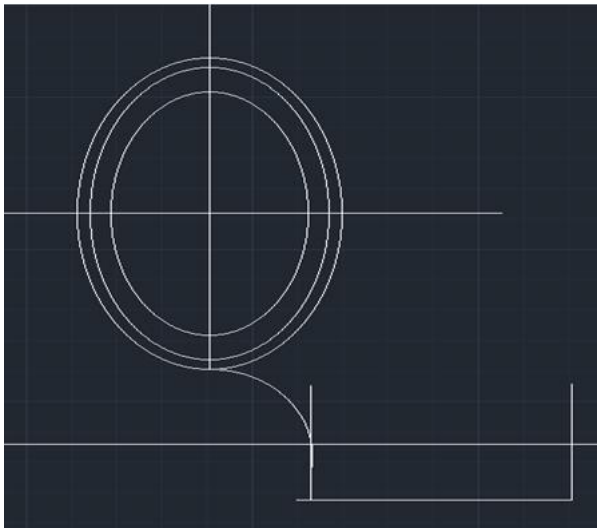


Figure : 02

3. Draw four 45 degree angle from x axis in each quadrant and draw two ellipse in each quadrant.

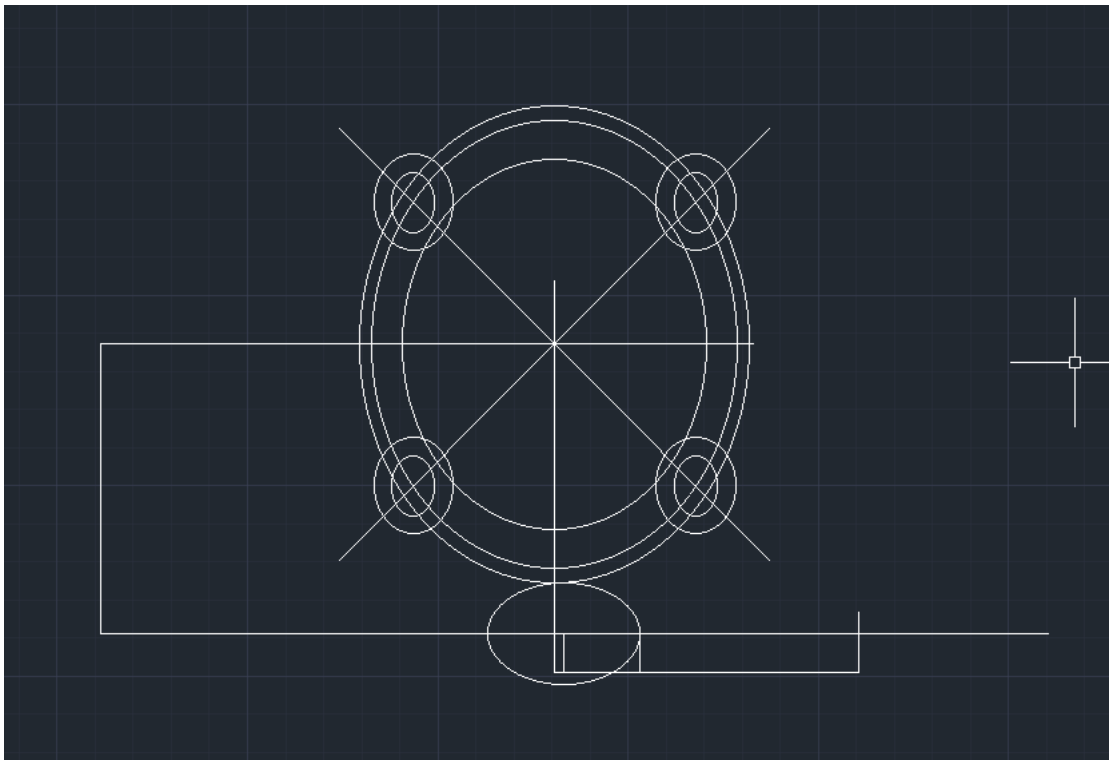


Figure : 03

4. Trim the unnecessary lines.

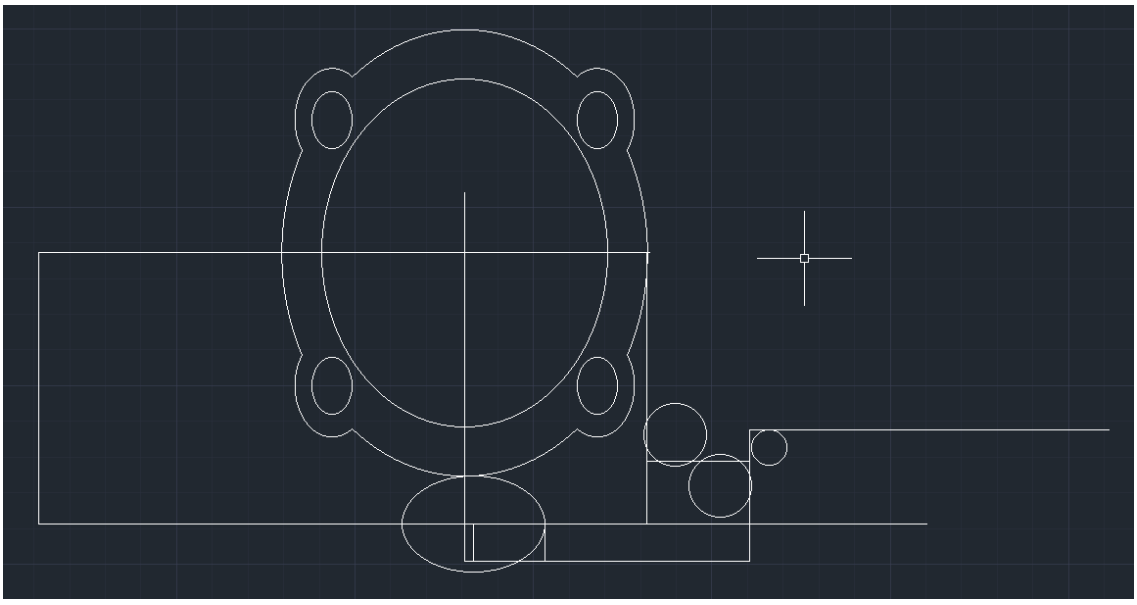


Figure : 04

5. Draw rest of the ellipse determining actual distance from given figure and trim unnecessary lines.

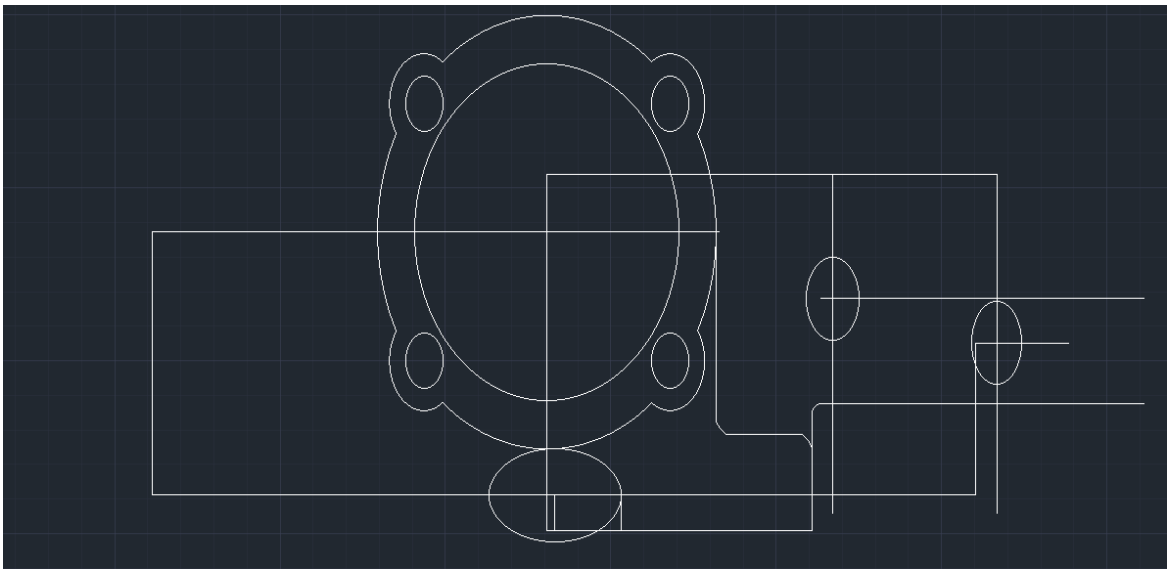


Figure : 05

6. Complete all ellipse then draw the tangent of the ellipse and connect together.

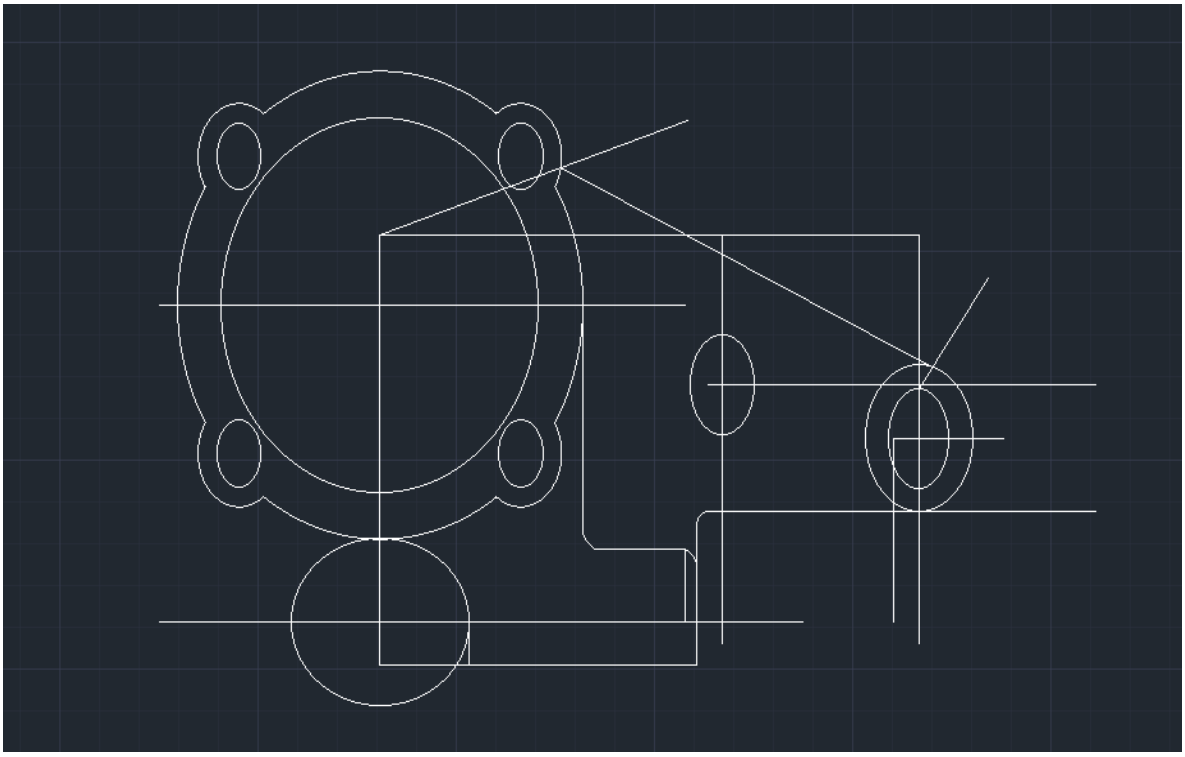


Figure : 06

7. Now trim all unnecessary lines from the figure.

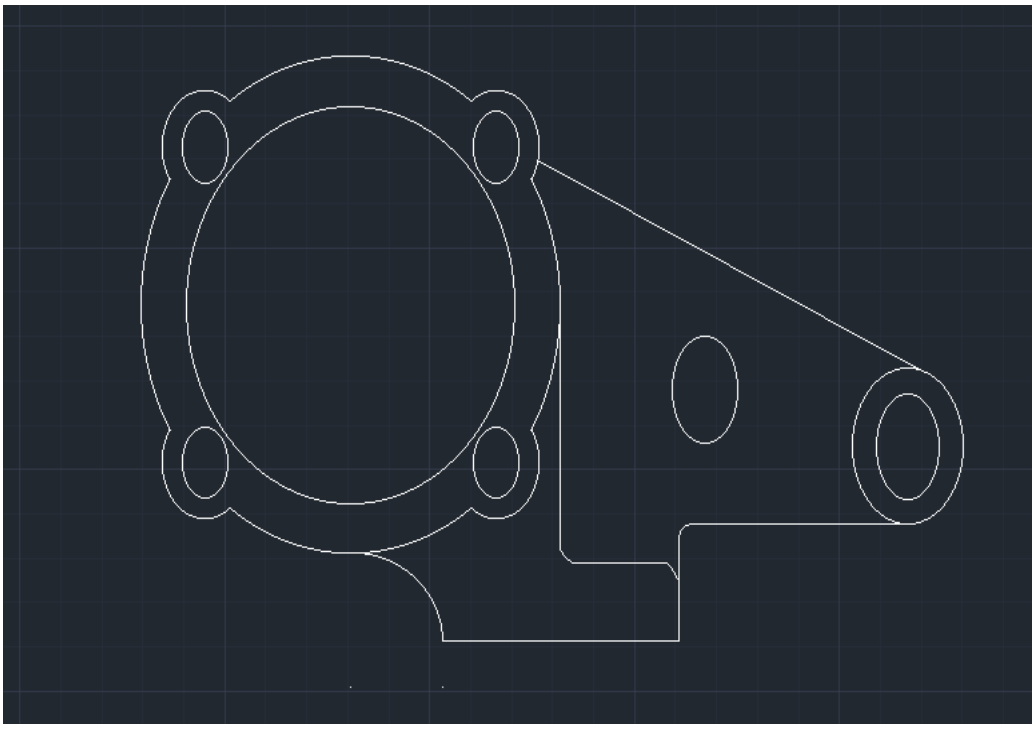


Figure : 07

6. FINAL OUTPUT

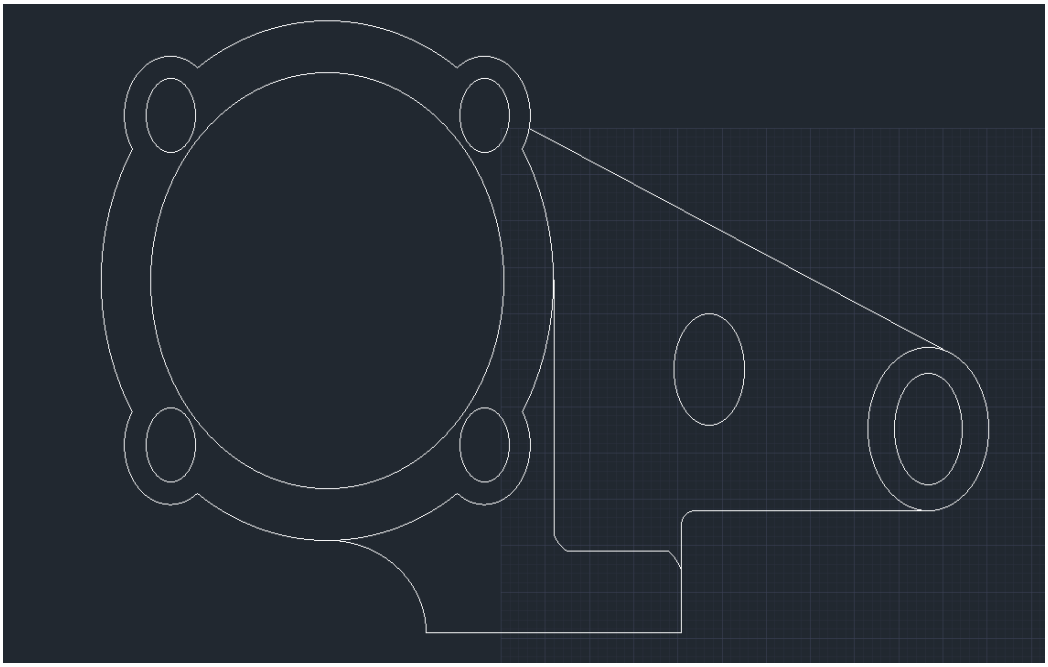


Figure : 08

7. ANALYSIS AND DISCUSSION

1. This experiment helps us to learn about basic Auto Cad operations such as create lines, circle, rectangle, ellipse, etc.
2. In this experiment I was facing some trouble while drawing the angle in each quadrant. When I give 45 degree as angle input in 2nd, 3rd or 4th quadrant from X axis, each time the line get fixed in 1st quadrant. When I put 45, 135, 225, 315 degree in particular quadrant then it works fine.
3. By this experiment we obtain the knowledge about 2D drawing technique and we can implement these tools to implement real life object.